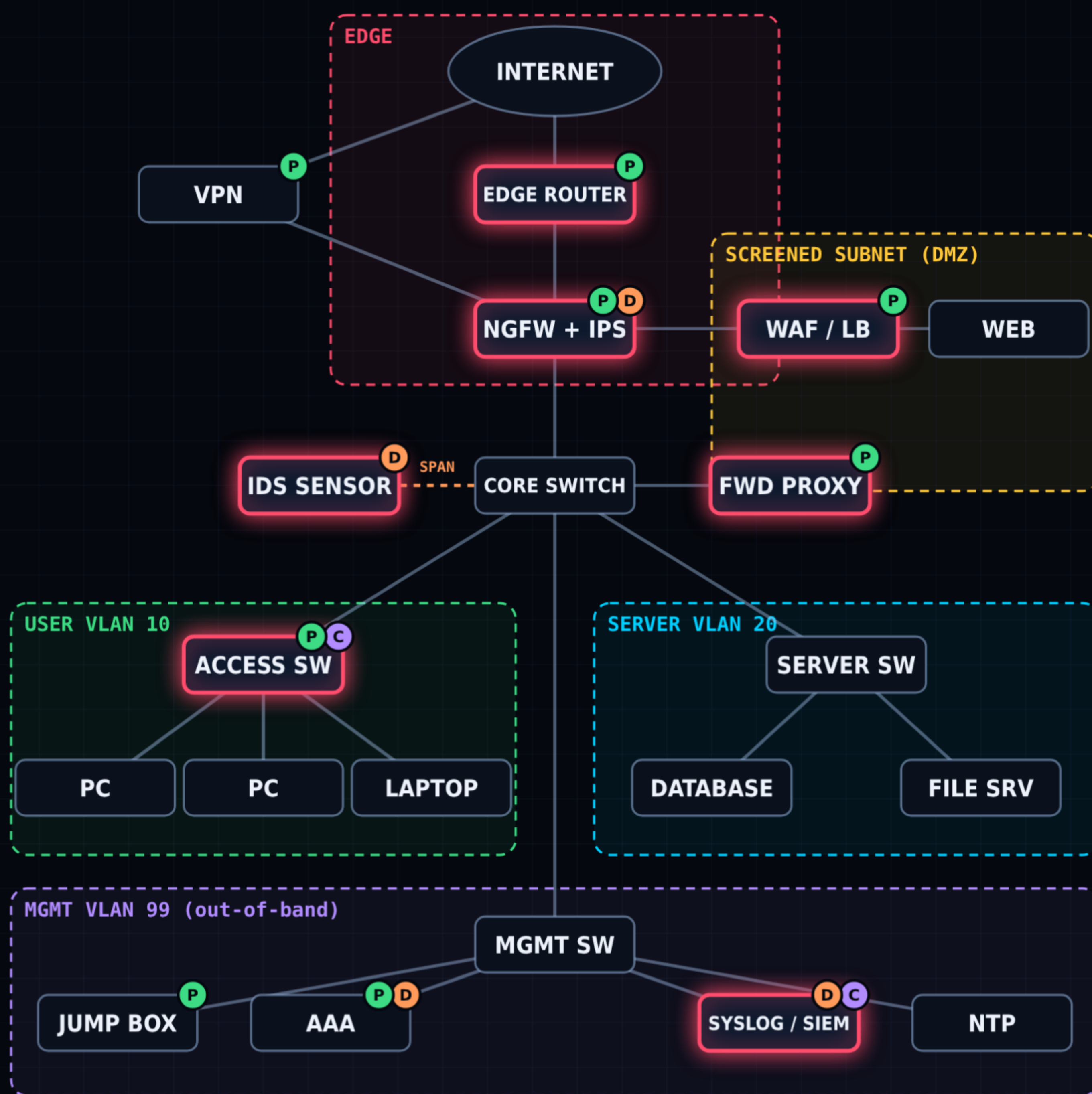


I MAPPED EVERY WALL A HACKER HITS ON THE WAY IN.



- Where each security control **actually** goes
- Why it goes **there**
- How to build it in **Cisco Packet Tracer**

SWIPE →

FIRST, THE 3 JOBS

EVERY CONTROL DOES ONE OF THREE JOBS.



Preventive: stops it

Firewall rule, ACL, 802.1X, WAF. The door is locked.



Detective: sees it

IDS, SIEM, logs. The camera is recording.

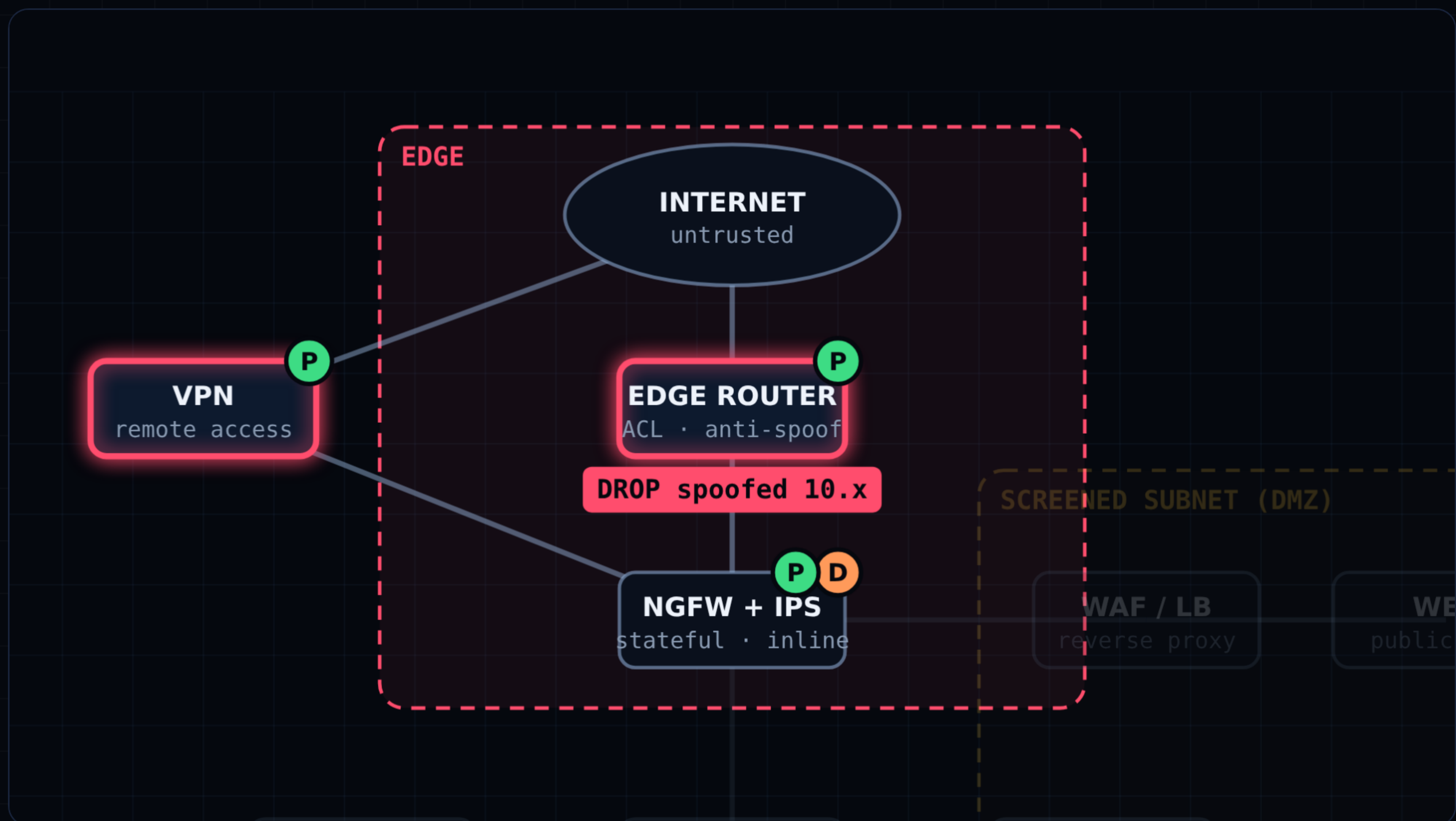


Corrective: fixes it

Quarantine VLAN, auto-isolate, restore from backup. Someone changes the locks.

Weak networks pick one. Strong networks layer all three. Watch for the P / D / C tags on every slide.

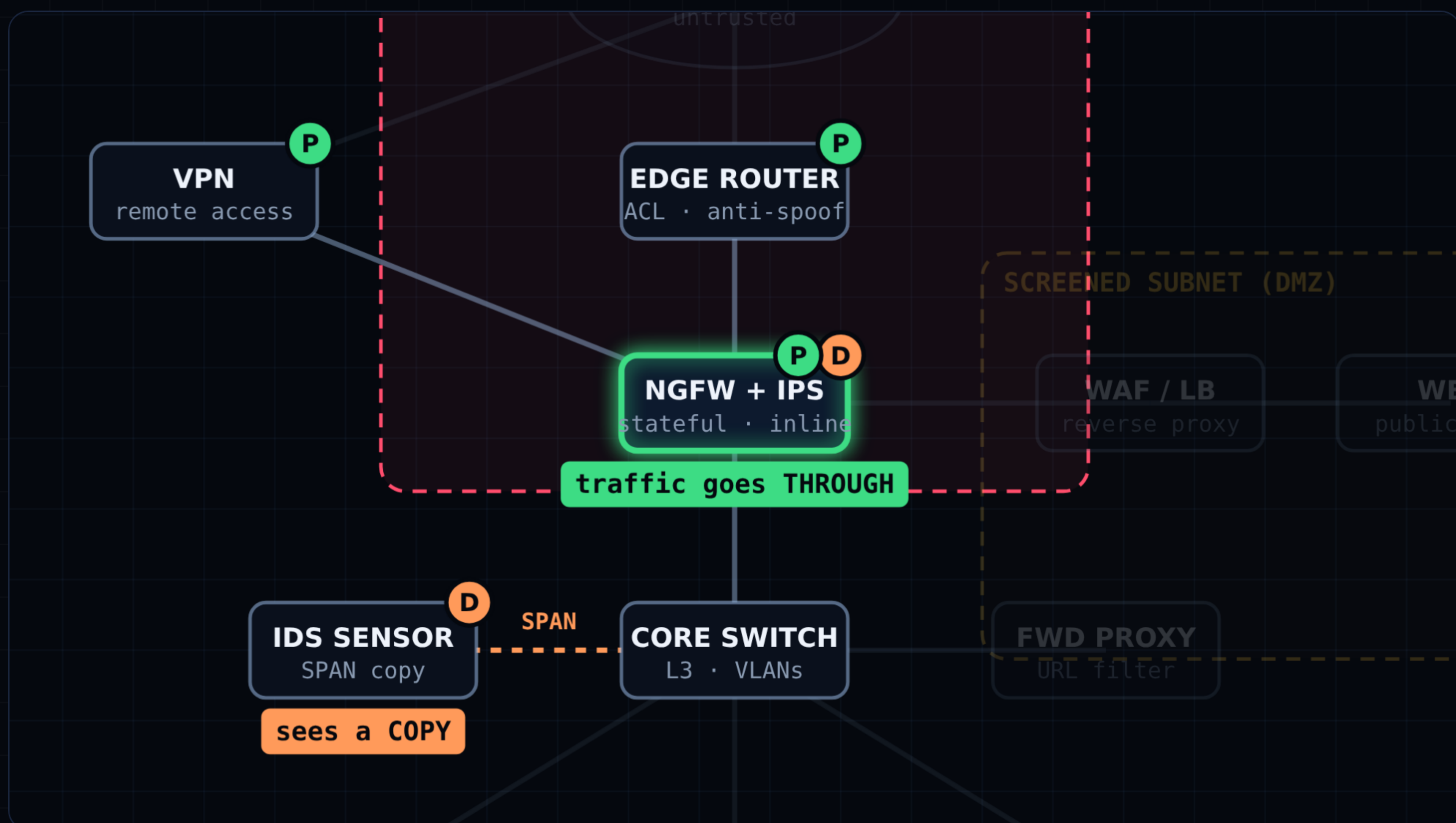
FILTER EARLY. FILTER CHEAP.



- The **edge router ACL** drops obvious junk first: packets from outside claiming to be **private IPs**, ports you never use.
- **VPN** lands remote users **at the firewall**, not straight inside.

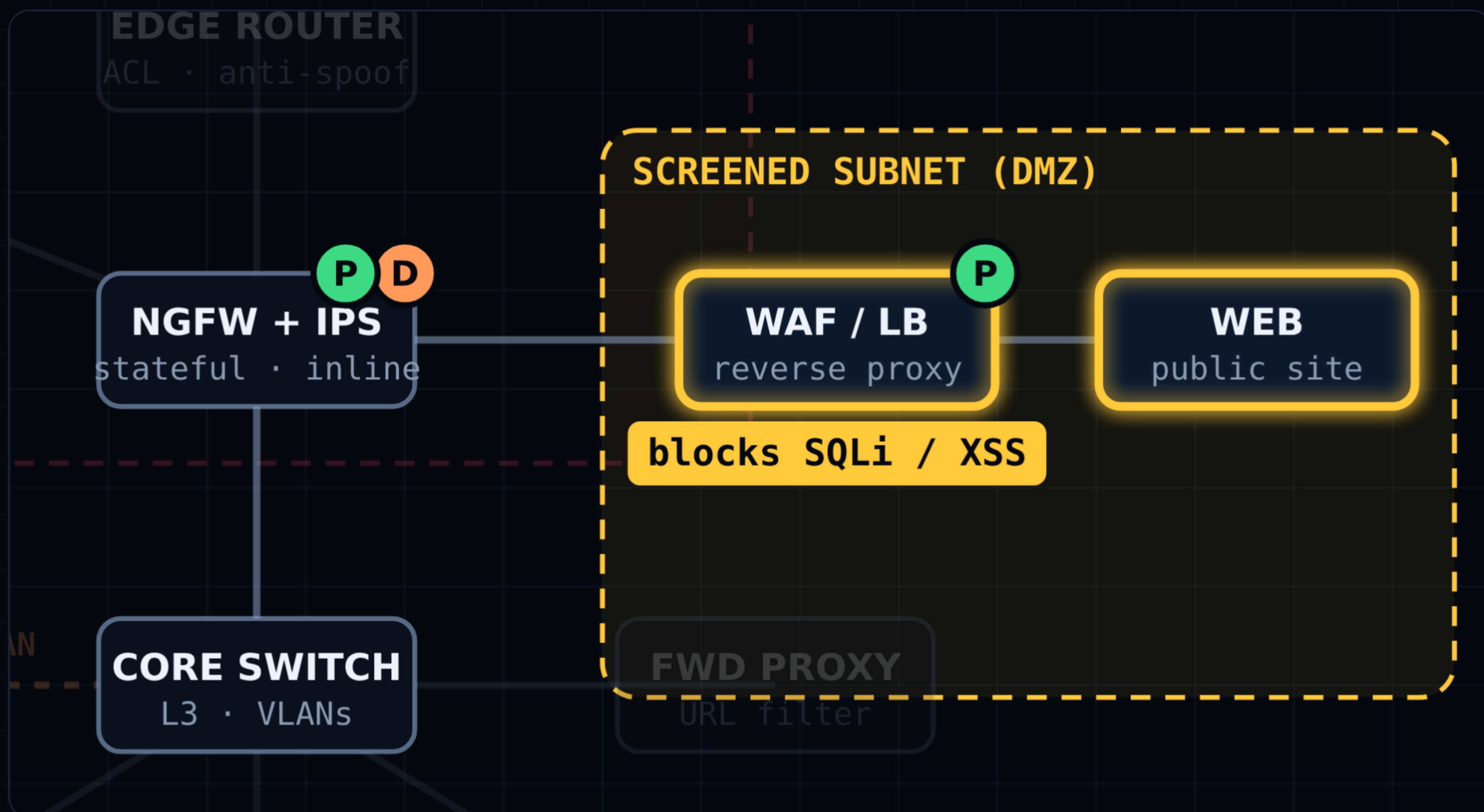
Cheap filter before the expensive one.

INLINE MEANS IT CAN SAY NO.



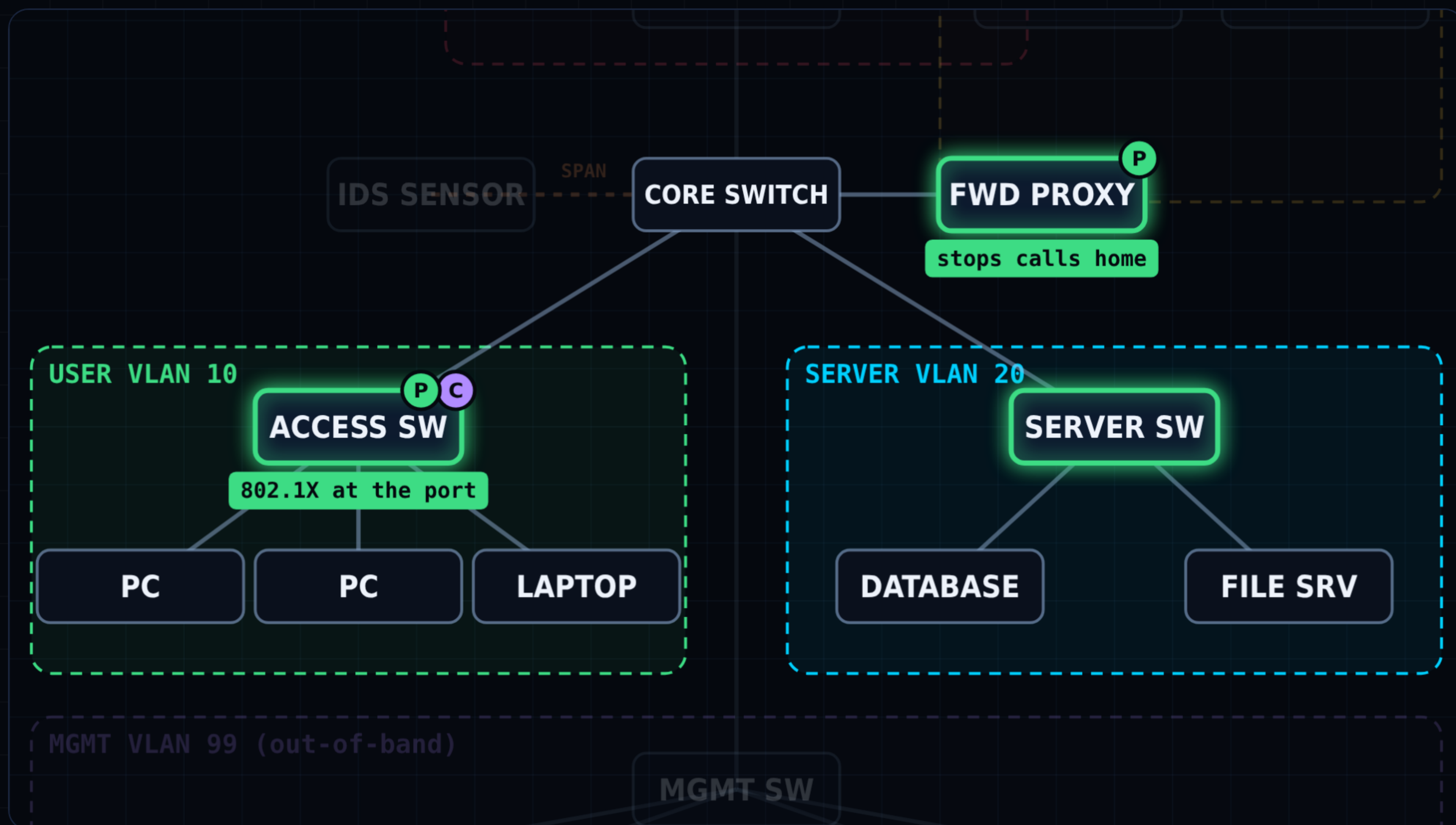
- **IPS = inline.** Traffic passes through it, so it can block.
- **IDS = passive.** It watches a SPAN/TAP copy. It alerts, it can't block.
- If the inline box dies: **fail-closed** (security wins) or **fail-open** (uptime wins)?

PUBLIC STUFF NEVER SITS INSIDE.



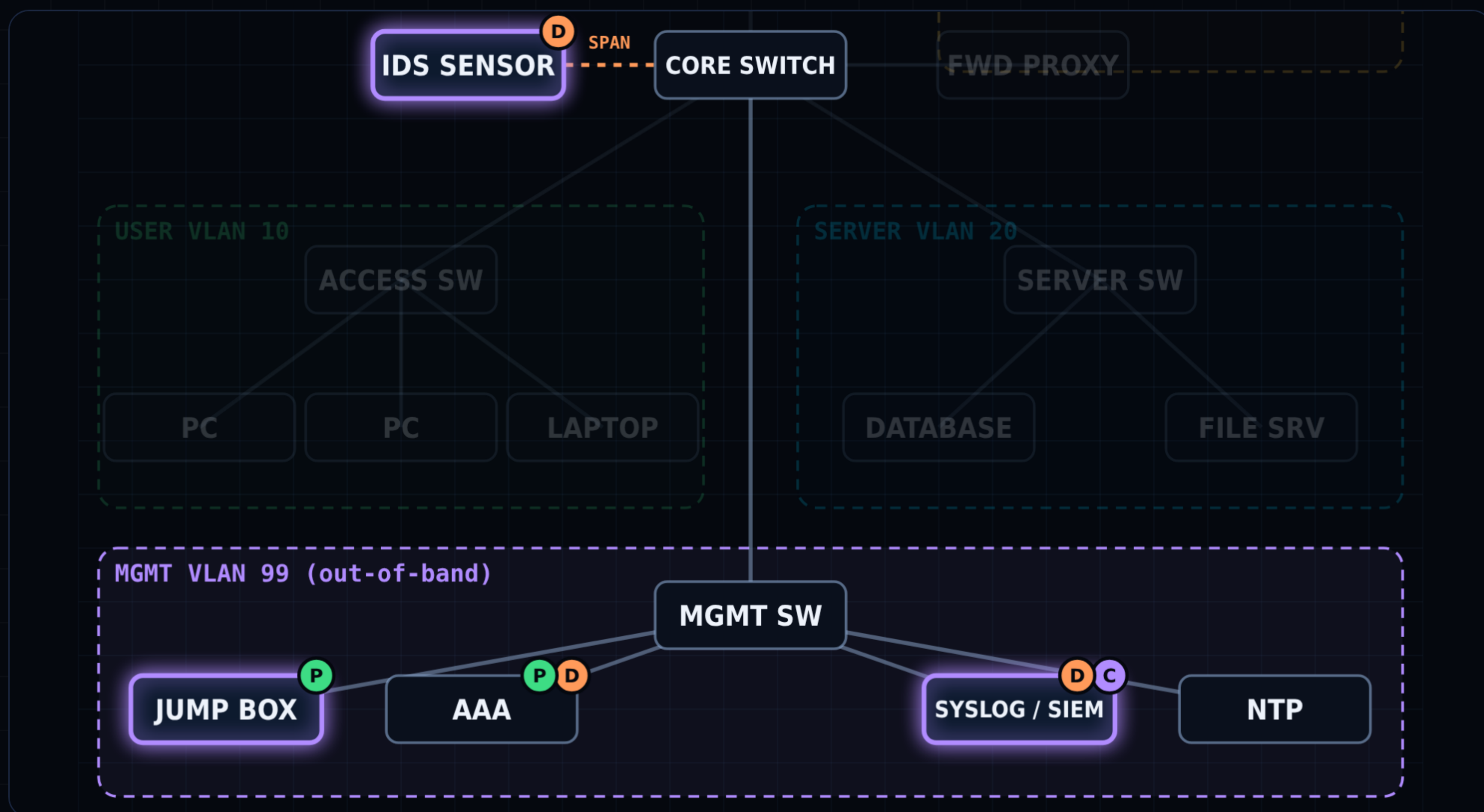
- Anything the internet can reach goes in the **DMZ**.
- The **WAF** reads web requests, not just ports.
- If the web server gets owned, the attacker is **stuck in the DMZ**, not in your network.

NO ID, NO NETWORK.



- **NAC / 802.1X** checks devices at the switch port. Unknown laptop goes to a quarantine VLAN.
- **VLANs + ACLs** keep users away from servers.
- The **forward proxy** controls what goes **out** (malware calling home).

DETECTION NEEDS ITS OWN LANE.



- **IDS + SIEM** collect and correlate. SIEM can also **auto-isolate** a host (corrective).
- **Mgmt VLAN:** admins only get in through the **jump box**, logged by **AAA**.
- **NTP** matters: if clocks disagree, you can't rebuild the timeline.

REAL STORY · TARGET, 2013

ONE VENDOR LOGIN. 40M CARDS.

STEP 1 Attackers phish an **HVAC contractor** and steal its login to Target's vendor portal.

STEP 2 From there they move across the network to the **checkout registers**. Vendor access wasn't walled off.

STEP 3 Card-stealing malware runs on registers. Security tools **raise alerts**, but nobody acts in time.

RESULT About **40 million cards** stolen, plus data on up to 70 million people.

Two fixes: **segment vendor access** (preventive) and **make sure someone answers the alerts** (detective needs a human).



Green line = where a segment wall should have stopped them.

YOUR LAB, **LAYER BY LAYER.**

Edge ACL

ISR router: `ip access-list extended EDGE-IN`

Firewall

ASA 5506-X: `nameif + security-level`

IDS

Sniffer device on the core link

NAC stand-in

2960 switch: `port-security`

AAA · Syslog · NTP

Server device, turn on each service

```
! drop spoofed private IPs at the edge
deny ip 10.0.0.0 0.255.255.255 any
deny ip 192.168.0.0 0.0.255.255 any
permit ip any any
! one device per access port
switchport port-security maximum 1
switchport port-security violation shutdown
```

Full step-by-step guide on my GitHub → [signal-and-noise/labs](#)

Try it: switch Packet Tracer to **Simulation mode**, send a ping from outside with a 10.x source, and watch the edge router drop it.

TRADER'S TRANSLATION

DEFENSE IN DEPTH = RISK MANAGEMENT.

NETWORK

Edge ACL drops junk

=

TRADING

Pre-trade checklist skips bad setups

Segmentation / VLANs

=

Position size per trade

IDS / SIEM alerts

=

Price alerts + trade journal

Auto-isolate a host

=

Stop-loss + daily loss limit

No single layer is perfect. That's the point. You stack them so one miss isn't the end.

Small business, one budget: which layer do you buy first?

01

Comment your pick

02

Save for your next lab

03

Follow for the next layer